

DR. GORDON GUOCHENG QIAN

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BIO

Senior Research Scientist at Snap Inc. leading R&D on multimodal generation across image, video, and 3D. Builds end-to-end systems spanning large-scale data curation, model training, evaluation, and product deployment. Authored **24 top-tier publications (3.5K+ citations)**, filed **6 patents**, and shipped **3 research techniques** across Snapchat products including Dreams and AI Lens for **900M+** monthly active users. Also serves as **Area Chairs for ICLR 2026 and NeurIPS 2026**.

PROFESSIONAL EXPERIENCE

Snap Inc.

Palo Alto, CA, USA

Senior Research Scientist

Dec 2023 – Present

- Topic: **Media Generation (Video, Image, 3D), Personalization.**
- End-to-end R&D for media generation, including large-scale data curation, model training, evaluation, and product deployment. Built the first human-centric dataset with 30M+ images, 6M+ identities, the first data visualization platform to check data quality, the first streamable data loader, the first FSDP cross-node training infra, and the first evaluation leaderboard for the personalization research team.
- 2025/11-present: state-of-the-art video editing *EasyV2V* (CVPR 2026, 1 patent); state-of-the-art video alignment fine-tuning *Diffusion-DRF*; efficient video generation (WIP);
- 2023/12-2025/11: *Omni-ID*, *ComposeMe*, and *SR-Training* integrated into Snapchat products (Dreams, AI Lens such as “JoinMe”, “AddMe”), Canvas-to-Image (1 patent, Ad Gen prototype); 5 patents filed.

HIGHLIGHTED PROJECTS

Multimodal Image Generation

- Canvas-to-Image: Under review, unified multimodal control, 1 patent, Ad Gen prototype
- Omni-ID: CVPR 2025, first generative face representation, state-of-the-art personalized T2I generation, products integrated, 1 patent

Video Editing and Alignment

- EasyV2V: CVPR 2026, state-of-the-art video editing, 1 patent
- Diffusion-DRF: Under review, state-of-the-art video alignment fine-tuning

Image-to-3D Generation

- Magic123: ICLR 2024, novel hybrid score distillation, 450+ citations, 1.6K+ GitHub stars

Scalable Spatial Understanding

- PointNeXt: NeurIPS 2022, 34% accuracy gain, 1200+ citations, 1K+ GitHub stars
- ASSANet: NeurIPS 2021 Spotlight, 54× speed-up, 120+ citations

INTERNSHIP EXPERIENCE

Snap Inc.

May 2023 – Sep 2023

- Project: Image/text-to-3D generation

Meta Platforms

Aug 2022 – Jan 2023

- Project: Efficient backbone for long-range video understanding

Microsoft Research Asia

Dec 2021 – May 2022

- Project: Generalizable 3D geometric representation learning

Megvii Technology Research

Jul 2021 – Nov 2021

- Project: Tree-search based efficient neural architecture search

EDUCATION

King Abdullah University of Science and Technology (KAUST)

Jeddah, Saudi Arabia

Ph.D. in Computer Science, specializing in 3D Vision & Generation

May 2019 - Dec 2023

*CEMSE Research Excellence Award (Top 10 among all CEMSE students)***Xi'an Jiaotong University (XJTU)**

Xi'an, China

Bachelor's degree in Engineering with honors

Aug 2014 - Jul 2018

Outstanding Undergraduate (Top 10 among all undergraduates)

PUBLICATIONS

See google scholar for the full list: <https://scholar.google.com/citations?user=DUDaxg4AAAAJ>
 Citations: 3572; H-index: 19; First-authored/corresponding top-tier conference/journal papers: 9; Top-tier conference/journal papers: 19. (Date: Mar 27, 2026) Selected publications are listed below. Here, * denotes co-first authorship, † denotes corresponding author.

- [1] Yusuf Dalva*, **Guocheng Qian***[†], Maya Goldenberg, Tsai-Shien Chen, Kfir Aberman, Sergey Tulyakov, Pinar Yanardag, and Kuan-Chieh Jackson Wang. Canvas-to-Image: Compositional image generation with multimodal controls. *under review*, 2025 [paper] [project]
- [2] **Guocheng Qian**, Kuan-Chieh Wang, Or Patashnik, Negin Heravi, Daniil Ostashev, Sergey Tulyakov, Daniel Cohen-Or, and Kfir Aberman. Omni-ID: Holistic identity representation designed for generative tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 [paper] [project]
- [3] **Guocheng Qian**, Daniil Ostashev, Egor Nemchinov, Sergey Tulyakov, Kuan-Chieh Jackson Wang, and Kfir Aberman. ComposeMe: Attribute-specific image prompts for controllable human image generation. *SIGGRAPH Asia*, 2025 [paper] [project]
- [4] Yifan Wang, Yanyu Li, **Gordon Guocheng Qian**[†], Sergey Tulyakov, Yun Fu, and Anil Kag. Diffusion-drif: Free, rich, and differentiable reward for video diffusion fine-tuning. *under review*, 2026 [paper] [project]
- [5] Anujraaj Argo Goyal, **Guocheng Qian**[†], Huseyin Coskun, Aarush Gupta, Himmy Tam, Daniil Ostashev, Ju Hu, Dhritiman Sagar, Sergey Tulyakov, Kfir Aberman, and Kuan-Chieh Jackson Wang[†]. Preventing shortcuts in adapter training via providing the shortcuts. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025 [paper] [project]
- [6] **Guocheng Qian**, Jinjie Mai, Abdullah Hamdi, Jian Ren, Aliaksandr Siarohin, Bing Li, Hsin-Ying Lee, Ivan Skorokhodov, Peter Wonka, Sergey Tulyakov, and Bernard Ghanem. Magic123: One image to high-quality 3d object generation using diffusion priors. In *International Conference on Learning Representations (ICLR)*, 2024 [paper] [project]  450  1.6k
- [7] **Guocheng Qian**, Yuchen Li, Houwen Peng, Jinjie Mai, Hasan Hammoud, Mohamed Elhoseiny, and Bernard Ghanem. PointNeXt: Revisiting pointnet++ with improved training and scaling strategies. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 23192–23204, 2022 [paper]  1125  980
- [8] **Guocheng Qian**, Hasan Hammoud, Guohao Li, Ali Thabet, and Bernard Ghanem. ASSANet: An anisotropic separable set abstraction for efficient point cloud representation learning. In *Advances in Neural Information Processing Systems (NeurIPS Spotlight)*, volume 34, pages 28119–28130, 2021 [paper]
- [9] **Guocheng Qian**, Abdulellah Abualshour, Guohao Li, Ali Thabet, and Bernard Ghanem. PU-GCN: Point cloud upsampling using graph convolutional networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11683–11692, June 2021 [paper]  322  194  120
- [10] **Guocheng Qian**, Xingdi Zhang, Abdullah Hamdi, and Bernard Ghanem. Pix4Point: Image pretrained transformers for 3d point cloud understanding. In *International Conference on 3D Vision (3DV)*, 2024 [paper]

- [11] **Guocheng Qian**, Yuanhao Wang, Jinjin Gu, Chao Dong, Wolfgang Heidrich, Bernard Ghanem, and Jimmy S Ren. Rethinking learning-based demosaicing, denoising, and super-resolution pipeline. In *IEEE International Conference on Computational Photography (ICCP)*, pages 1–12. IEEE, 2022 [paper]  Stars 298
- [12] **Guocheng Qian**, Ruihang Zhang, Tsai-Shien Chen, Yusuf Dalva, Anujraaj Goyal, Willi Menapace, Ivan Skorokhodov, Daniil Ostashev, Meng Dong, Arpit Sahni, Ju Hu, Sergey Tulyakov, and Kuan-Chieh Jackson Wang. LayerComposer: Multi-human personalized generation via layered canvas. *Under review*, 2025 [paper] [project]
- [13] Thuan Hoang Nguyen, Jiahao Luo, Hao Li, Jian Wang[†], and **Guocheng Qian**[†]. FFAvatar: Few-shot, feedforward, and generalizable avatar reconstruction. *under review*, 2025
- [14] G. Li*, Matthias Müller*, **Guocheng Qian***, Itzel C. Delgadillo, Abdullellah Abualshour, Ali K. Thabet, and Bernard Ghanem. DeepGCNs: Making gcns go as deep as cnns. *IEEE transactions on pattern analysis and machine intelligence (T-PAMI)*, 45(6):6923–6939, 2023 [paper] [project]  Stars 629
- [15] G. Li*, **Guocheng Qian***, Itzel C. Delgadillo*, Matthias Müller, Ali K. Thabet, and Bernard Ghanem. SGAS: Sequential greedy architecture search. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1617–1627, 2020 [paper] [project]  Cites 279  Stars 163
- [16] Hanwen Liang, Junli Cao, Vidit Goel, **Guocheng Qian**, Sergei Korolev, Demetri Terzopoulos, Konstantinos Plataniotis, Sergey Tulyakov, and Jian Ren. WonderLand: Navigating 3d scenes from a single image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 [paper] [project]
- [17] Sherwin Bahmani, Ivan Skorokhodov, **Guocheng Qian**, Aliaksandr Siarohin, Willi Menapace, Andrea Tagliasacchi, David B. Lindell, and Sergey Tulyakov. AC3D: Analyzing and improving 3d **Camera Control in Video Diffusion** transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 [paper] [project]
- [18] Sherwin Bahmani, Ivan Skorokhodov, Aliaksandr Siarohin, Willi Menapace, **Guocheng Qian**, Michael Vasilkovsky, Hsin-Ying Lee, Chaoyang Wang, Jiaxu Zou, Andrea Tagliasacchi, David B. Lindell, and Sergey Tulyakov. VD3D: Taming large video diffusion transformers for 3d camera control. In *International Conference on Learning Representations (ICLR)*, 2025 [paper] [project]
- [19] Zhenxing Mi, Kuan-Chieh Wang, **Guocheng Qian**, Hanrong Ye, Runtao Liu, Sergey Tulyakov, Kfir Aberman, and Dan Xu. I think, therefore i diffuse: Enabling multimodal in-context reasoning in diffusion models. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025 [paper] [project]
- [20] Tuan Duc Ngo, Ashkan Mirzaei, **Guocheng Qian**, Hanwen Liang, Chuang Gan, Evangelos Kalogerakis, Peter Wonka, and Chaoyang Wang. DELTA_v2: Accelerating dense 3d tracking. In *Under review*, 2025 [paper] [project]
- [21] Abdullah Hamdi, Luke Melas-Kyriazi, Jinjie Mai, **Guocheng Qian**, Ruoshi Liu, Carl Vondrick, Bernard Ghanem, and Andrea Vedaldi. GES: Generalized exponential splatting for efficient radiance field rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024 [paper]
- [22] Chen Zhao, Shuming Liu, Karttikeya Mangalam, **Guocheng Qian**, Fatimah Zohra, Abdulmohsen Alghannam, Jitendra Malik, and Bernard Ghanem. Dr2Net: Dynamic reversible dual-residual networks for memory-efficient finetuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024 [paper]
- [23] Jinjie Mai, Wenxuan Zhu, Sara Rojas Martinez, Jesus Zarzar, Abdullah Hamdi, **Guocheng Qian**, Bing Li, Silvio Giancola, and Bernard Ghanem. Tracknerf: Bundle adjusting nerf from sparse and noisy views via feature tracks. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024 [paper] [project]
- [24] Tianyu Zhang, **Guocheng Qian**, Jin Xie, and Jian Yang. Fastpci: Motion-structure guided fast point cloud frame interpolation. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024 [paper]
- [25] Guohao Li, Chenxin Xiong, **Guocheng Qian**, Ali Thabet, and Bernard Ghanem. DeeperGCN: Training deeper gcns with generalized aggregation functions. *IEEE transactions on pattern analysis and machine intelligence (T-PAMI)*, 2023 [project]  Stars 629

- [26] Jun Chen, Deyao Zhu, **Guocheng Qian**, Bernard Ghanem, Zhicheng Yan, Chenchen Zhu, Fanyi Xiao, Mohamed Elhoseiny, and Sean Chang Culatana. Exploring open-vocabulary semantic segmentation without human labels. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023 [paper]

ACADEMIC EXPERIENCE

Teaching Assistant Deep Learning for Visual Computing (2020 - 2022), Design and Analysis of Algorithms (2021)

Area Chair ICLR 2026, NeurIPS 2026

Reviewer CVPR, ECCV, ICCV, ICML, NeurIPS, SIGGRAPH, SIGGRAPH Asia, AAAI, IJCAI, T-PAMI, TVCG

Conference Organizer Tutorial on Graph Machine Learning for Computer Vision @ CVPR'22 , Workshop on Learning 3D with Multi-View Supervision @ CVPR'23&CVPR'24&CVPR'26

HONORS & AWARDS

KAUST CEMSE Research Excellence Award Top 12/400

KAUST CEMSE Dean's List Award Top 20%

KAUST Fellowship Full tuition support, monthly stipend, housing, and medical coverage

National First-Class Scholarship Top 1% nationwide

Outstanding Undergraduate Award Top 10/2000

Excellent Student Cadre Top 3%

TECHNICAL SKILLS

- **Multimodal Representation, Understanding, and Generation:** 24 top-tier publications across representation learning, spatial understanding, multimodal generation, and alignment fine-tuning
- **PyTorch & Open-source:** Magic123 🌟 >1.6K+ stars, PointNeXt 🌟 1K+ stars, DeepGCNs 🌟 1.2K+ stars
- **Large-scale Systems:** end-to-end multimodal pipelines covering large-scale data collection, annotation, processing, and distributed training on up to 256 GPUs; data visualization platform to check data quality; benchmark leaderboard for model evaluation.
- **End-to-end R&D:** shipped 3 research techniques in Snapchat products including Dreams and AI Lens